

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 110

ZERO-THEORY MATERIAL VERIFICATION TRANSMISSION PROTOCOLS

No whiteboard needed — bench-provable modules only
hot-swap valves, zero-skate templates, prism-locks



understood before the coffee cools

PARALLEL SWAP · DRILL TEMPLATE · PRISM-LOCK
IF IT NEEDS A WHITEBOARD, IT DOES NOT SHIP

Module-selection system — v1.0 in force

GP-IM-110

Revision v1.0 — 23 August 2026

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VETTED BATCH 17 — PASS · TRANSIENT FIGURE OWED

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 110

PHASE 110: ZERO-THEORY MATERIAL VERIFICATION TRANSMISSION PROTOCOLS — STATUS: CURRENT — WORDING FLAG SEATED. No whiteboard required. Phase 110 selects the modules a shop foreman grasps before his coffee cools: the Phase 70/87 twin-track parallel conduit hot-swap (make-before-break cooling transfer, valves shifting flow between pristine and service lines), the Phase 82 zero-skate drill templates (a drill bushing married to a command strip), and the Phase 95 prism-lock anchor grid (30-degree angled leverage replacing expansion bolts, zero epoxy). The selection criterion was 'zero theoretical' — every item demonstrable on a bench in one gesture. One honest flag rides with it: the '0% pressure variance' phrase took the batch-17 wording treatment — a real valve move has transients, and the owed figure is a measured allowable pressure transient, not a literal zero. The physics note already says it: nothing in this module needs a whiteboard.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-110, v1.0 — 23 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the zero-theory law as seated (contents line, the module bodies, the physics note, the origin — verbatim) — §2 why each module is debate-proof (graded) — §3 the system in the selections — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 110: **Zero-Theory Material Verification Transmission Protocols.** Coordinates outbound documentation delivery profiles to target practical, floor-level operational bottlenecks, forcing immediate workshop trial initializations by prioritizing raw mechanical fixes over theoretical fluff.'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Technical Comprehension via Practical Floor-Level Physics '

Industrial Target: CASC Strategic Blueprint Deployment & Zero-Theory Modules

THE COMPLIANCE COOLING & HYDROCONDUIT SELECTION (MODULE C) (VERBATIM)

- **System Infrastructure:** Packages the Phase 70/87 Twin-Track Parallel Conduit Grid for reactor thermal management.
- **Pressure-Equilibrium Logic:** Establishes the 0% pressure-variance hot-swap protocol, driven via mechanical dual-gated selector valves that shift flow between pristine and service lines at the same millisecond.
- **Operational Value:** Eliminates reactor thermal shut-down risks, providing combat puncture isolation and backflow prevention under fire while engines run at peak capacity.

THE INDUSTRIAL DRILLING & PRISM-LOCK BALANCING (MODULE D) (VERBATIM)

- **System Tooling Profile:** Packages the Phase 82 Zero-Skate Target Templates for precision drilling on sensitive aerospace alloys.
- **Mechanical Bit Capture:** Outlines the use of elastomeric pull-tab adhesive templates with reinforced aperture guide rings to completely halt drill-bit skating and surface stress scoring.
- **Lever-Clinch Anchor Geometry:** Integrates the 30-degree inward triangular pin matrix, replacing destructive expansion bolts with a removable structural foundation grid requiring zero epoxy.

PHYSICS NOTE — PHASE 110 (KIMI AUDIT: AFFIRMED — ALL FOUR MODULES ARE ZERO-THEORY) (VERBATIM)

- **The selection criterion was 'no debate possible,' and all four pass.** The parallel fluid swap is make-before-break transfer — standard practice in power distribution and chemical plants, here applied to cooling with a flat volumetric cross-section so pressure never blips. The drill template is a drill bushing married to a command strip; any machinist who has watched a bit walk across titanium understands it in one second. The prism-lock replaces sideways expansion force (which cracks rock) with angled leverage (which tightens under load) — the same wedge physics as a rock-climbing nut. Nothing in this module needs a whiteboard.

ORIGIN OF THOUGHT — PHASE 110 (VERBATIM)

Archer, 18 July: "what other 2 would you recommend that is zero theorhetical they will quickly understand" — and then, dictating the cover brief in his own plain-English shock-value register:

"Obsolescence of Reusable rocket replacement technology for cargo deployment. High speed kilos per minute graphene production. Ending line sludge with zero variance parallel fluid swaps. Replacement of deep rock Dynabolt Anchor. would be my intro, plain english at shock value."

Why it matters, in plain English: the last two modules were chosen by one test — could a shop foreman grasp it before his coffee cools? A valve that swaps a dirty pipe for a clean one without stopping the reactor. A sticker that stops a drill bit from skating. Three angled pins that out-hold a Dynabolt and leave the rock unbroken. Archer's one-line titles do more work than a hundred-page proposal: each one names a pain the reader personally pays for, and ends the sentence before their attention can wander.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

MAKE-BEFORE-BREAK TRANSFER, AFFIRMED: the parallel hot-swap is standard practice in power distribution and chemical plants — dual-gated selector valves shift flow between pristine and service lines without interrupting delivery. Applied to reactor cooling with a flat volumetric cross-section, the pressure barely blips. 'Barely' is the honest word — see the flag below.

THE 0% FLAG, SEATED: the vet applied the same treatment given to '100%' and 'microsecond' — a mechanical valve system cannot be assumed to hold literally zero pressure variation: valve motion brings pressure waves, flow acceleration, restriction, line elasticity, fluid compressibility, control response. The design can make the transient very small; the text cannot claim literal zero. Owed: a measured allowable transient (§9).

THE DRILL TEMPLATE, AFFIRMED: elastomeric pull-tab adhesive plus a reinforced aperture guide ring is a drill bushing married to a command strip — any machinist who has watched a bit walk across titanium understands it in one second. Zero-skate is geometry, not theory.

THE PRISM-LOCK, AFFIRMED: 30-degree inward triangular pins replace sideways expansion force (which cracks rock) with angled leverage (which tightens under load) — the same wedge physics as a rock-climbing nut, removable, zero epoxy. Ordinary mechanical concepts, correctly assembled.

§3 — THE SYSTEM IN THE SELECTIONS

- THE SWAP: Phase 70/87 parallel conduits — pristine and service lines, dual-gated mechanical selector valves, flow shifted without shutdown.
- THE GATE: the hot-swap protocol — pressure transient owed as a measured figure (the '0%' phrase flagged by vet 17).
- THE TEMPLATE: Phase 82 zero-skate drill guides — adhesive elastomer, reinforced aperture rings, no bit wander on aerospace alloys.

- THE LOCK: Phase 95 prism-locks — 30-degree angled pin matrix; leverage instead of expansion; the rock unbroken.
- THE CRITERION: zero-theory — every selection demonstrable on a bench in one gesture; comprehension before coffee cools.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE ZERO-THEORY DOCTRINE (seated with the selection): if a foreman needs a whiteboard, it does not ship — the module is the proof.
- THE PLAIN-ENGLISH-SHOCK-VALUE DOCTRINE (the origin): the one-line titles name a pain the reader personally pays for — 'Ending line sludge with zero variance parallel fluid swaps' — and end before attention wanders.
- THE KILL-THE-ABSOLUTE DOCTRINE (batch-17 treatment): '0% variance' joins 'frictionless' and '100%' on the wording ledger — measurable figures outrank adjectives.
- THE BENCH-OVER-BOARD DOCTRINE (the physics note): nothing in this module needs a whiteboard — the file's recurring test for what counts as proven.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Bench the swap: run the dual-gated selector between pristine and service lines at full flow — film the transfer.
- STEP 2 — Measure the transient: pressure taps both sides of the valve; publish the actual allowable transient figure (§9's owed number).
- STEP 3 — Stick the template: adhesive drill guide on alloy plate; drill the pattern; gauge the holes — zero skate or back to the bench.
- STEP 4 — Set the pins: prism-lock matrix into test rock; load to spec; confirm removal leaves the rock unbroken.
- STEP 5 — Package the proofs: each module one gesture, one film, one page — zero-theory as shipped.
- STEP 6 — Publish the ledger: transient figure, hole gauges, pull-out curves — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 70/87 — the parallel conduit grid (GP-IM-070/087): the swap's plumbing.
- PHASE 82 — the zero-skate templates (GP-IM-082): the drilling module.
- PHASE 95 — the prism-locks (GP-IM-095): the anchor module.
- PHASE 109 — the tranche frameworks (GP-IM-109): the packaging these selections ride in.
- PHASE 122 — the calibration authority (GP-IM-122): the same bench-governs-the-number law, written for tolerances.

§8 — DO-NOT-BUILD

- DO NOT print '0% pressure variance' as a claim — the measured transient figure is the law until the bench says otherwise (flag seated).
- DO NOT ship a module that needs a whiteboard — zero-theory is the selection criterion, not a slogan.
- DO NOT expansion-bolt the rock — the prism-lock exists because sideways force cracks the substrate.
- DO NOT swap lines without the make-before-break gate — an interrupted reactor cooling line is the failure this phase deletes.

§9 — OWED

OWED: the measured allowable pressure transient across the hot-swap valve event — pressure-tap data from the bench, published as the replacement for the '0% variance' phrase (the design may make the transient very small, the text cannot claim literal zero). Nothing else outstanding.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-110, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). Built 23 August 2026, third of the 20-chapter 'ok next 20' shift (the author's order, 23 August 2026: 'ok next 20'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566 and 572): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for

the day when i get back' / 'ok next 20'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 110